

A Parent-Aware Computer-Assisted Lower Bound for Packing 17 Unit Squares in a Square

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<https://github.com/Mira-acc/17squares>

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Abstract

Let $s(n)$ be the least side length of a square containing n unit squares with pairwise disjoint interiors and arbitrary orientations. We prove

$$s(17) > \frac{46129999999859}{9997499999900} = 4.614153538416645696808 \dots$$

The proof assigns a finite nonnegative measure to the container and selects a smaller closed square inside each possible packed square. An exact finite calculation shows that every selected square captures more than one seventeenth of the total mass. Following the parent-aware reduction of Guzhou0806, we check only centers attainable by the enclosing square. We then sharpen the bound by moving the selected squares within their parents while reusing the verified coverage. As a separate result, we construct intersection-trigger charges and give an exact 29-square example that separates them from nonnegative spatial measures.

1 Introduction

Write $Q_L = [-L/2, L/2]^2$ for $L > 0$. For a positive integer n , let $s(n)$ be the minimum L for which Q_L contains n unit squares with pairwise disjoint interiors. Boundary contacts are allowed, and each square may rotate independently. The minimum is attained; a compactness argument is given below. For $n = 17$, Bidwell's construction gives

$$s(17) \leq L_B, \quad L_B = 4.6755300936045509516 \dots,$$

as recorded in Friedman's survey and Ellsworth's catalogue [1, 4]. We establish the following lower bound.

Theorem 1. *There is no packing of seventeen unit squares in a square of side*

$$S_* = \frac{46129999999859}{9997499999900}.$$

Consequently, $s(17) > S_$.*

The proof uses a finite nonnegative measure. Inside every possible packed square, called a *parent*, we select a smaller closed square, called a *core*. The cores lie strictly inside their parents, so cores selected from a packing are disjoint even when the parents touch. We arrange that every selected core captures mass at least γ , while the total mass is less than 17γ . Seventeen parents would therefore require more mass than is available.

The difficulty is to prove the mass bound for every translation and orientation of a parent. A finite catalogue assigns one core shape to each interval of parent orientations. For that interval we check all centers where a parent can occur. This domain can be smaller than the set of centers where the core alone fits, and thus imposes fewer coverage constraints.

Contributions and dependencies. Weighted covering extends the classical unavoidable-point argument; recent constructions and exact verification methods include those of Burns, Massaccesi, and Levy [7, 9, 11]. We use Guzhou0806’s R012 parent-angle catalogue and legal-center reduction [12], whose antecedents include Levy’s parent-center restrictions. Our base certificate refines the angle catalogue and changes both the weights and the spatial support, giving

$$s(17) > \frac{18452}{3999} = 4.61415353838459614903\dots$$

A recentering argument then proves Theorem 1, improving this base constant by approximately 3.205×10^{-11} without another translation sweep. The intersection-trigger result is independent of this numerical bound: it shows a strict advantage over spatial measures on a specified finite family, not coverage of every possible parent.

Sections 2–3 give the certificate criterion, the complete finite reduction, and the base computation. Section 4 proves the sharpening, and Section 5 treats triggers. Reproduction instructions and supplementary results, including the dated history of improvements, are in the appendices. The computational claims use rational data and exact arithmetic; the geometric reductions are proved here. The complete argument has not been formalized in a proof assistant.

2 Parent-aware certificates

2.1 Counting disjoint cores

A *charge* assigns a nonnegative number $C(K)$ to each closed core K . Its *budget* is any number $\mathcal{B}$ such that

$$\sum_i C(K_i) \leq \mathcal{B}$$

for every finite pairwise-disjoint family of cores. For a finite nonnegative measure μ , the charge $C(K) = \mu(K)$ has budget $M = \mu(\mathbb{R}^2)$.

Lemma 2 (Disjoint-core counting). *Suppose every admissible parent admits a closed core strictly inside its interior with charge at least $\gamma > 0$. If the charge has budget $\mathcal{B} < n\gamma$, then n admissible parents cannot have pairwise disjoint interiors.*

Proof. Choose one such core from each parent. These cores are disjoint, so their total charge is both at least $n\gamma$ and at most $\mathcal{B}$, a contradiction. $\square$

The strict containment is essential for atomic measures: an atom on a shared parent boundary must not be counted twice. The numerical certificate below uses only point masses. The more general language will also apply to the charges in Section 5.

2.2 Orientations, containment, and legal centers

Let D_4 be the eight symmetries of the container. We choose a D_4 -invariant measure, so an individual parent may be reflected into orientation $\phi \in [0, \pi/4]$ without changing any core mass. The square’s

own quarter-turn symmetry accounts for the remaining orientations. This reduction imposes no symmetry on a hypothetical packing.

Use the half-angle parameter $u = \tan(\phi/2)$, whose physical folded range is $[0, \sqrt{2} - 1]$. For the rational computations it is convenient to define, more generally for $0 \leq u < 1$,

$$c(u) = \frac{1 - u^2}{1 + u^2}, \quad s(u) = \frac{2u}{1 + u^2}, \quad f(u) = c(u) + s(u). \quad (1)$$

The first two quantities are the cosine and sine of $2 \arctan u$; $f(u)$ is the axis-parallel width of a unit square in that direction.

A catalogue row consists of an interval $I_j = [\alpha_j, \beta_j] \subset [0, 1)$, a core parameter $t_j \in [0, 1)$, and a core side $B_j > 0$. The parent side is $A > 0$. To compare the two directions, define

$$g(t, u) = c(t)c(u) + s(t)s(u) + |s(t)c(u) - c(t)s(u)|. \quad (2)$$

When the relative angle is at most $\pi/4$ in absolute value, $B_j g(t_j, u)$ is the width of the core across either parent normal.

Lemma 3 (Interval containment). *Suppose the angles represented by t_j and each endpoint of I_j differ by at most $\pi/4$ in absolute value. Put*

$$G_j = \max\{g(t_j, \alpha_j), g(t_j, \beta_j)\}.$$

If $B_j G_j < A$, then the concentric closed core lies strictly inside the side- A parent for every $u \in I_j$.

Proof. The largest absolute angular difference from a fixed direction occurs at an interval endpoint. On $[0, \pi/4]$, the function $\cos v + \sin v$ is increasing. Thus the core width across either parent normal is at most $B_j G_j < A$. Both normal-coordinate inequalities are strict, proving the claim. $\square$

The relative-angle hypothesis is checked rationally at each endpoint: the dot product of the direction vectors must be positive and at least the absolute value of their cross product. No sampling assertion is used between endpoints.

A side- A parent with parameter u fits in Q_L precisely when its center lies in the square of half-side $(L - Af(u))/2$, provided this is nonnegative. The complete envelope for an interval has a simple exact description.

Lemma 4 (Legal-center envelope). *Set*

$$m_j = \min\{f(\alpha_j), f(\beta_j)\}, \quad d_j = \frac{L - Am_j}{2}.$$

If $d_j \geq 0$, the union of the legal center sets over I_j is exactly $D_j = [-d_j, d_j]^2$.

Proof. Since

$$f'(u) = \frac{2(1 - 2u - u^2)}{(1 + u^2)^2},$$

f has a single interior maximum and no interior minimum on $[0, 1)$. Its minimum on I_j is therefore m_j . The nonempty center squares are nested, and the endpoint attaining m_j supplies the largest one. $\square$

In particular, every point of D_j is attained by a legal parent at an endpoint orientation. When an interval extends slightly beyond the folded physical range, using its full envelope merely checks additional centers.

2.3 The certificate criterion

Theorem 5 (Parent-aware certificate). *Fix a positive integer n and $L, A > 0$. Let μ be a finite nonnegative D_4 -invariant measure supported in Q_L , and let $\gamma > 0$. Suppose a finite catalogue has the following properties:*

- (i) *Its intervals cover $[0, \sqrt{2} - 1]$.*
- (ii) *Every row satisfies Lemma 3, and its domain D_j from Lemma 4 has $d_j > 0$.*
- (iii) *Every core from row j centered in D_j has mass at least γ .*
- (iv) *$M := \mu(\mathbb{R}^2) < n\gamma$.*

Then n side- A squares cannot pack in Q_L , and $s(n) > L/A$.

Proof. Fold each parent separately and choose a catalogue interval containing its orientation. Its center lies in D_j , so the specified concentric core has mass at least γ and lies strictly inside the parent. Apply the inverse reflection to the core. Invariance of μ preserves its mass, and strict containment is preserved. Lemma 2 excludes a packing.

Scaling by A excludes a unit-square packing at side L/A . To obtain the strict inequality, note that the minimum defining $s(n)$ is attained. A grid packing supplies a finite upper bound on the container side; centers and orientations then range over compact sets. Containment is closed under limits. Interior overlap is an open condition on a pair of square poses, so interior nonoverlap is closed. A sequence of packings whose container sides approach the infimum therefore has a feasible limit. Since the endpoint L/A has been excluded, that minimum is strictly larger. $\square$

The remaining universal condition is (iii). We next reduce it to a finite arrangement calculation and prove that the calculation includes all boundary placements.

3 Finite verification and the base bound

3.1 Capture rectangles

For the computation, write $\mu = \sum_{k=1}^N \omega_k \delta_{p_k}$ with $\omega_k \geq 0$. Assume L, A , the catalogue parameters, the atom coordinates, and the weights are rational. Fix one row and abbreviate its parameters by t, B, d . Write $c = c(t) > 0$ and $s = s(t) \geq 0$. Rotate a potential core center (x, y) into the core frame:

$$\xi = cx + sy, \quad \zeta = -sx + cy.$$

For a spatial atom p , let (ξ_p, ζ_p) be its rotated coordinates. The closed core captures p exactly when its center belongs to

$$R_p = [\xi_p - B/2, \xi_p + B/2] \times [\zeta_p - B/2, \zeta_p + B/2]. \quad (3)$$

Thus the mass at a center is the sum of weights of the rectangles containing it. The legal-center square becomes the rotated square

$$\Omega = \{(\xi, \zeta) : |c\xi - s\zeta| \leq d, |s\xi + c\zeta| \leq d\}. \quad (4)$$

All rectangle edges and domain vertices are rational.

3.2 Which cells must be checked?

Split the ξ -axis at every vertical edge of a capture rectangle and at the domain extremes $\pm d(c + s)$. Between consecutive events, the rectangles active in the ξ direction are fixed. Split the ζ -axis at all horizontal rectangle edges. Each resulting open rectangular cell has a constant captured mass. We must identify exactly those cells meeting the interior of Ω .

For an open vertical slab (a, b) inside the domain's horizontal extent, consider the projection of $\Omega \cap ([a, b] \times \mathbb{R})$ onto the ζ -axis. Convexity makes this a closed interval $[\ell, h]$. The projection of the corresponding open interior intersection is (ℓ, h) : the closed intersection is the closure of its interior, and linear projection of that nonempty convex interior is an open interval. Consequently a horizontal cell (v_k, v_{k+1}) is reachable with positive area exactly when

$$\max\{v_k, \ell\} < \min\{v_{k+1}, h\}. \quad (5)$$

It is the projection of the *whole slab* that matters; a cross-section at its midpoint can miss reachable cells.

Here is an exact formula for that projection. If $s > 0$, the lower and upper boundaries of Ω are

$$\begin{aligned} l(\xi) &= \max\{(c\xi - d)/s, (-d - s\xi)/c\}, \\ h_0(\xi) &= \min\{(c\xi + d)/s, (d - s\xi)/c\}. \end{aligned}$$

The convex function l is minimized at $d(c - s)$, and the concave function h_0 is maximized at $-d(c - s)$. Define $\text{clamp}(z; a, b) = \min\{b, \max\{a, z\}\}$. Then

$$\ell = l(\text{clamp}(d(c - s); a, b)), \quad h = h_0(\text{clamp}(-d(c - s); a, b)). \quad (6)$$

When $s = 0$, the domain is axis aligned and $[\ell, h] = [-d, d]$. These formulas also handle a domain vertex lying inside the slab.

The finite algorithm sums the active weights on every horizontal cell satisfying (5), for every vertical slab. Its smallest sum is the smallest mass attained away from the arrangement and domain boundaries. All comparisons use rational arithmetic.

Lemma 6 (Boundary completeness). *For nonnegative atomic weights and $d > 0$, the minimum over the reachable open cells equals the minimum over the whole closed domain Ω .*

Proof. Every point of Ω is a limit of interior points avoiding the finitely many rectangle edges. A subsequence lies in one fixed open cell. Every capture rectangle containing that cell is closed, so it also contains the limit point. Nonnegative weights imply that the mass at the limit is at least the mass in that cell. Conversely, each reachable open cell contains a point of Ω . The two minima are therefore equal. $\square$

This proves both obligations of the finite reduction: every tested mass is realized by a legal center, and no untested placement has smaller mass. The implementation additionally reconstructs a rational center for each reported minimum and recounts all atoms there. That recount checks the reported witness; the arrangement argument above establishes exhaustiveness.

3.3 The verified certificate

Proposition 7 (Computational base certificate). *For*

$$L = \frac{4613}{1000}, \quad A = \frac{3999}{4000},$$

the accompanying rational data satisfy Theorem 5 with $n = 17$ and

$$M = \frac{4249997911}{250000000} = 16.999991644, \quad \gamma = \frac{100000003}{100000000} = 1.000000030.$$

In particular, $s(17) > 18452/3999$.

Computer-assisted verification. The measure has 3,280 atoms in 417 positive D_4 -orbits. Its 4,391 catalogue intervals meet exactly, start at zero, and end at the rational value $207107/500000 > \sqrt{2} - 1$. Each row satisfies the endpoint containment checks and has $d_j > 0$. The exact sweep just described verifies mass at least γ throughout every D_j . Finally,

$$17\gamma - M = \frac{4433}{500000000} > 0.$$

Theorem 5 gives the conclusion. Appendix A identifies the data, replay procedure, and arithmetic assumptions. $\square$

4 Reusing coverage by recentering cores

Concentricity helped construct the base certificate, but the counting proof requires only strict containment. We can therefore shrink the parents and container slightly while moving each chosen core back into a center domain where its mass has already been verified. The measure stays fixed in the common container-centered coordinate system.

4.1 A general transfer principle

Let K_j denote a core shape centered at the origin, and suppose every translate $y + K_j$ with $y \in D_j$ has charge at least γ . For a parent shape P centered at the origin, define

$$E_j(P) = \{v : v + K_j \subset \text{int } P\}.$$

A parent centered at x can use entry j exactly when some $y \in D_j$ satisfies $y - x \in E_j(P)$. Equivalently, $x \in D_j - E_j(P)$, where $D - E = \{y - v : y \in D, v \in E\}$. For centrally symmetric P and K_j , the set $E_j(P)$ is also centrally symmetric, so $D_j - E_j(P) = D_j + E_j(P)$.

Proposition 8 (Recentered-core transfer). *Suppose C has budget $\mathcal{B} < n\gamma$. If every admissible parent $P + x$ satisfies*

$$x \in \bigcup_j (D_j - E_j(P)),$$

then n such parents cannot have pairwise disjoint interiors.

Proof. Membership supplies an index j and a certified center y with $y + K_j \subset \text{int}(P + x)$. The selected core has charge at least γ . Apply Lemma 2. $\square$

No continuity of the atomic mass under translation is assumed: the mass statement is used only at an already verified center.

4.2 An exact criterion for square domains

Let a new parent have side $a > 0$, orientation $\phi \in [0, \pi/4]$, and container side $H > 0$. Define

$$F(\phi) = \cos \phi + \sin \phi, \quad q = \frac{H - aF(\phi)}{2}.$$

Thus its legal center set is $[-q, q]^2$ when $q \geq 0$, and $f(u) = F(2 \arctan u)$ in the earlier notation. Suppose the old core has side B , angle θ , and certified center domain $[-d, d]^2$. Its width across either parent normal is

$$w(\phi) = B(|\cos(\theta - \phi)| + |\sin(\theta - \phi)|).$$

Write $[z]_+ = \max\{z, 0\}$.

Lemma 9 (Square recentering). *If $q \geq 0$ and $d \geq 0$, every legal parent center admits a strictly interior core with center in $[-d, d]^2$ if and only if*

$$w(\phi) + F(\phi)[H - aF(\phi) - 2d]_+ < a. \quad (7)$$

Proof. Clamp each coordinate of a parent center to $[-d, d]$. Each coordinate moves by at most $[q - d]_+$, so displacement along either parent normal is at most $F(\phi)[q - d]_+$. Adding the core half-width proves sufficiency.

For necessity when $q > d$, choose the parent center (q, q) . Every available core center has both coordinates at most d , so displacement along the normal $(\cos \phi, \sin \phi)$ has magnitude at least $F(\phi)(q - d)$. Strict containment forces (7). When $q \leq d$, concentric placement is available, and the necessary and sufficient condition is $w(\phi) < a$. $\square$

4.3 The scalar sharpening

For row j of the base certificate, retain m_j, G_j, d_j and set

$$\eta_j = A - B_j G_j > 0.$$

Choose $0 < \varepsilon < A$ and $0 < \delta < L$, and put $a = A - \varepsilon$, $H = L - \delta$. Let F_j be an upper bound for $f(u)$ on I_j . For any folded physical parameter $u \in I_j$,

$$\begin{aligned} H - af(u) - 2d_j &= A(m_j - f(u)) + \varepsilon f(u) - \delta \\ &\leq \varepsilon F_j - \delta. \end{aligned}$$

Moreover, $w(2 \arctan u) = B_j g(t_j, u) \leq A - \eta_j$. Substitution into (7) shows that the sufficient condition

$$\eta_j > \varepsilon + F_j[\varepsilon F_j - \delta]_+ \quad (8)$$

allows every legal new parent in that interval to use its old core and center domain. Empty legal-center sets impose no obligation.

Proof of Theorem 1. Take

$$\varepsilon = 10^{-11}, \quad \delta = \frac{141}{10^{13}}.$$

For each interval not containing $\sqrt{2} - 1$, use the larger endpoint value of f as F_j . For an interval containing that maximum, use the rational upper bound $1414213563/10^9 > \sqrt{2}$. Exact evaluation of all 4,391 inequalities (8) gives

$$\min_j (\eta_j - \varepsilon - F_j[\varepsilon F_j - \delta]_+) > 10^{-15}.$$

The finite arithmetic check is included in the reproduction procedure.

Each new parent therefore admits a core whose mass is at least the same γ as in Proposition 7. Fold and unfold each parent as before; both containers have the same symmetry center, and the old measure remains D_4 invariant. The old total mass M is still a valid budget, even if some of its atoms lie outside the smaller container. Since $M < 17\gamma$, Proposition 8 excludes seventeen side- a parents in Q_H . Scaling and the compactness argument give

$$s(17) > \frac{H}{a} = \frac{L - \delta}{A - \varepsilon} = \frac{46129999999859}{9997499999900}.$$

□

5 Intersection-trigger charges

The preceding bound uses additive spatial mass. We now construct charges that also reflect incompatibility between possible square poses. This section gives a general budget rule and a finite separation result; a packing improvement would additionally require coverage for every legal parent.

5.1 Intersecting triggers and their budget

Let $T_1, \dots, T_r$ be nonempty finite sets of points, called triggers, with $T_i \cap T_j \neq \emptyset$ for $i \neq j$. Define

$$C_T(K) = \mathbf{1}\{T_i \subseteq K \text{ for some } i\}.$$

Proposition 10. *The charge C_T has budget one on pairwise-disjoint cores.*

Proof. If two cores activate the same trigger, they share a point because that trigger is nonempty. If they activate different triggers, they share a point in their intersection. Thus two disjoint cores cannot both be charged. □

For example, Levy's two-of-three threshold charge [11] uses the three two-element subsets of a three-point set as triggers. More generally, for a finite set X , integer $k \geq 1$, and weight $w \geq 0$, a threshold charge activates when a core contains at least k points of X . Its budget is $w\lfloor |X|/k \rfloor$, since disjoint cores have disjoint traces on X . For arbitrary nonempty triggers, their matching number—the largest number of pairwise-disjoint triggers—is a valid budget.

To obtain triggers from geometry, start with square poses $V_1, \dots, V_r$ having pairwise-overlapping interiors, with $r \geq 2$. Choose

$$p_{ij} = p_{ji} \in \text{int } V_i \cap \text{int } V_j, \quad T_i = \{p_{ij} : j \neq i\}.$$

Then $T_i \subset \text{int } V_i$ and $p_{ij} \in T_i \cap T_j$. The resulting budget-one charge activates on every V_i . Because the finitely many witnesses lie strictly inside their squares, activation persists for a sufficiently small change in each pose.

5.2 A finite separation from spatial measures

Proposition 11 (29-core separation). *There are 29 closed squares K_i of side $499/500$ and an intersection-trigger charge with budget one such that $C_T(K_i) = 1$ for every i , whereas every nonnegative Borel measure ν satisfying $\nu(K_i) \geq 1$ for all i has*

$$\nu(\mathbb{R}^2) \geq \frac{81407286808}{59431493389} = 1.3697668048682659 \dots > 1.$$

Computer-assisted construction and proof. The accompanying data specify 29 rational unit-square poses V_i , positive rational weights λ_i , and 406 pair witnesses. Every required witness has inward normal distance greater than $1/1000$ from the boundary of its unit square. It therefore lies inside the concentric closed square K_i of side $499/500$. The 29 triggers each contain 28 witnesses, and their pairwise intersections prove $C_T(K_i) = 1$ and budget one.

The exact weighted sum and maximum pointwise load are

$$W := \sum_i \lambda_i = \frac{81407286808}{79999999999}, \quad \Lambda := \max_{x \in \mathbb{R}^2} \sum_i \lambda_i \mathbf{1}_{\text{int } V_i}(x) = \frac{59431493389}{79999999999}.$$

To check the latter equality, partition the plane at all vertex abscissae and all edge-intersection abscissae. In each open vertical slab, edge order is fixed, so a rational cross-section enumerates every open cell's covering set. Coincident events are grouped. At any boundary point, all open squares containing that point also contain a neighborhood of it; hence no boundary load exceeds the open-cell maximum. A maximizing witness is recounted directly. The rational data and replay are specified in Appendix A.

Since $K_i \subset \text{int } V_i$, the same upper bound Λ holds for the weighted indicators of the closed cores, including their boundaries. For any finite measure ν as in the statement,

$$W \leq \sum_i \lambda_i \nu(K_i) = \int \sum_i \lambda_i \mathbf{1}_{K_i} d\nu \leq \Lambda \nu(\mathbb{R}^2).$$

Division by Λ proves the stated bound. Infinite total mass satisfies it trivially. $\square$

5.3 Coverage geometry and symmetry

At a fixed core direction, let (ξ_p, ζ_p) be the coordinates of a trigger point in the core frame. The centers of side- B cores containing all of T_i form the rectangle

$$R_i = \left[\max_{p \in T_i} \xi_p - B/2, \min_{p \in T_i} \xi_p + B/2 \right] \times \left[\max_{p \in T_i} \zeta_p - B/2, \min_{p \in T_i} \zeta_p + B/2 \right],$$

where a reversed interval means the rectangle is empty. Thus C_T is the indicator of $\bigcup_{i=1}^r R_i$. A finite vertical-slab decomposition again gives exact coverage cells. Rectangles belonging to a single trigger charge must be evaluated as a union: an overlapping pair contributes one, not two.

One can also start from an overlap graph G , choosing a shared interior witness for each edge and a private interior witness for each isolated vertex. Each vertex trigger contains its incident edge witnesses, or its private witness if the vertex is isolated. Disjoint charged cores must select distinct, nonadjacent vertices, so any proved upper bound b on the independence number of G gives a budget b .

Nonnegative sums of charges have the sum of their budgets, with no independence assumption. In particular,

$$C(K) = \mu(K) + \sum_h w_h C_{T(h)}(K), \quad \mathcal{B} = M + \sum_h w_h b_h$$

satisfies Lemma 2 for $w_h \geq 0$. The recentering proposition also applies once the necessary coverage is proved. To use the folded-angle certificate, however, the *total charge* must be D_4 invariant. Otherwise coverage must be checked over the full orientation range. Averaging $C(gK)$ over $g \in D_4$ preserves its budget and gives an invariant charge, but its coverage must be established for that averaged charge. The finite separation above does not use an orientation fold.

Acknowledgments. The direct mathematical and software dependencies are credited above and in the artifact documentation. The research, computation, and exposition were developed with substantial AI assistance under human direction. This is a computer-assisted working preprint; independent external review remains open.

A Reproducibility and arithmetic assumptions

The public artifact [5] contains the exact inputs, checkers, and result records. The paper uses centered coordinates; data stored in $[0, L]^2$ are translated by $(-L/2, -L/2)$ when interpreted in this notation. This common change of origin also applies to the smaller container in the recentering argument.

A.1 Packing certificate

The directory `certificates/lower_bound_4p614153/` contains `certificate.json`, the authoritative base input; `verify.py` and `parent_sweep.cpp`, the exact replay; and `sharpening.py`, the finite recentering check. An orbit row (x, y, w) expands into all distinct D_4 images of $(x/10^5, y/10^5)$, each with weight w/D , where D is the recorded weight denominator. A catalogue row stores the four rational values $(\alpha_j, \beta_j, t_j, B_j)$.

The loader rejects duplicate orbit sites, negative weights, inconsistent mass totals, incomplete interval coverage, failed strict-containment checks, and degenerate center domains. It also verifies that the cores fit throughout the supplied domains. The default input’s exact byte identity is checked against the SHA-256 recorded in the package manifest and result record; the mathematical coverage is then recomputed. From the repository root, run

```
python3 certificates/lower_bound_4p614153/verify.py \
  --output /tmp/n17-replay --upstream --direct
python3 certificates/lower_bound_4p614153/test_verify.py
python3 certificates/lower_bound_4p614153/check_diagnostics.py
```

The output directory must not already exist. Requirements are Python 3.10+, a C++17 compiler with OpenMP, and Boost headers. Numerical optimization and network access are not part of verification.

The current catalogue requires 27,918,671 center slabs. The two accumulation modes, a segment tree and a direct prefix calculation, must agree on every interval minimum. They share the geometric partition and do not constitute two independent geometric proofs. With `-upstream`, the command also reconstructs and checks all 2,925 R012 intervals from the pinned source in [12]; it does not run that source’s Python sweep. Earlier research runs of the source-distinct Python implementation are documented separately in the result record.

Geometric comparisons use unbounded integers or exact rational arithmetic. Integer weight accumulators are signed 64-bit values; validation bounds the total nonnegative integer weight by 10^{12} , below the accumulator limits. The checker requires a complete output for every catalogue row and verifies the counting inequality using the recomputed minima.

The recentering check verifies every inequality (8), the bound on $\sqrt{2}$, positivity of the new parameters, and nonempty legal-center domains. Full exact values of its minimum slack and the cap in Appendix B.2 are retained in `result.json` and in the replay’s `sharpening.result.json`. The support and catalogue were found by numerical search; their validity depends only on this exact replay and the reductions proved in the main text.

A.2 Trigger certificate

The directory `certificates/intersection_trigger_29/` contains `intersection-atom.json`, `clique-29-poses-with-weights.csv`, the pinned source data, and an exact Python checker. Run

```
python3 certificates/intersection_trigger_29/verify.py
python3 certificates/intersection_trigger_29/test_verify.py
```

Only Python 3.10+ and its standard library are needed. The checker validates rational unit rotations, containment of the poses in the side-4.67 box, every shared witness and inward margin, and all fractional weights. It computes the 29-pose load across 1,475 arrangement slabs, including a direct recount of a maximizing point. Source hashes identify the supplied artifacts; the separation proof checks the selected poses directly and assumes no load bound for the full source family.

B Historical and supplementary results

B.1 Selected lower-bound history

Table 1 records the results that shaped this work and nearby benchmarks. Classical unavoidable-point arguments lead to the integral counting rule; fractional measures and restricted parent-center domains provide the successive relaxations used here. Other geometric approaches include the work of Stromquist, Nagamochi, and MacIver [2, 3, 13]. Detailed attribution and date evidence are retained in `CONTRIBUTORS.md` and `paper/HISTORY-DATES.md`.

Table 1: Selected $n = 17$ lower bounds, ordered by bound rather than date. Dates identify publication or retained records, not necessarily discovery; Green’s is the reported communication year. Rounded values marked $\approx$ are only summaries of their source bounds.

Date	Contributor / source	Bound	Contribution
2000	Green, via Friedman [1]	≈ 4.445208	Sixteen unavoidable points.
2026-07-18	Brandwijk [8]	4.45	Exact point certificate.
2026-08-10; 2026-08-11	Mira [5]	4.450837; 4.468292	Exact pose subdivision; later triangle witnesses.
2026-08-11	Fort [6]	4.456575	Independent exact pose subdivision.
2026-08-06	Burns [7]	4.4811	Fractional atomic measure.
2026-08-21	Massaccesi [9]	4.5058	LP-designed weighted support.
2026-08-16	anabologyco-maker [10]	4.5705	Weighted certificate with a Lean finite-check layer.
2026-09-04	Levy [11]	4.59	Reusable exact event sweeps and coverage checks.
2026-09-07	Mira continuation [5]	≈ 4.607029 ; ≈ 4.613029	Support and angle refinement; exact dilation.
2026-09-19	Guzhou0806 / R012 [12]	≈ 4.613046	Parent-angle catalogue and legal-center coverage.
2026-09-19	This work	≈ 4.614154	New measure and catalogue; analytic recentering.

R012’s exact constant is $461300/99999$. Intermediate search reports reached $922600/199979$ with its measure fixed, and then $230650/49993$ by reweighting the same support. Their separate certificates are not retained here. The published replay covers R012, the final base certificate, and the recentering corollary.

An exact retained diagnostic explains one limit of changing only the core catalogue. At parent side 99989/100000 in the side-4.613 container, a legal parent has R012 mass 0.998982, below the mass required by its common-charge counting inequality. No inner core can have more mass than the parent. This obstructs that fixed measure at target 461300/99989, rather than bounding $s(17)$ or the general method. The final certificate changes the measure by reweighting inherited sites and adding wall-adjusted support.

B.2 A limit on reusing the fixed core domains

The recentering argument admits a simple upper bound on what the unchanged core shapes and certified domains alone can prove. For an axis-aligned parent, write $W_j = B_j f(t_j)$. Lemma 9 requires

$$a > W_j, \quad H < 2a + 2d_j - W_j.$$

These inequalities follow already by considering a corner parent center. Thus even if the choice of entry depends on that center, some entry must satisfy them. For $a > W_j$,

$$\frac{H}{a} < 2 + \frac{2d_j - W_j}{a} \leq \max \left\{ 2, 1 + \frac{2d_j}{W_j} \right\}.$$

Consequently any such reuse argument has

$$\frac{H}{a} \leq C_* := \max_j \max \left\{ 2, 1 + \frac{2d_j}{W_j} \right\}, \quad C_* \approx 4.614153538422734.$$

The exact rational maximum is checked in the scalar replay. This is a limit on those unchanged core/domain statements, not an upper bound on $s(17)$ or on certificates using new shapes, domains, measures, or charges.

B.3 Additional trigger activations

The trigger checker also verifies 71 unit-square activations from the pinned source, of which 68 remain valid after shrinking to side 499/500. The 68 retained weights sum to 96836866527/79999999999. A separate exact arrangement calculation on the 71 unit squares gives maximum pointwise load 73959042636/79999999999 < 1. The integration argument used in Proposition 11 therefore verifies the earlier, weaker mass lower bound 96836866527/79999999999 > 1.21 on the 68 cores. The 29-core result in the main text is stronger. Neither calculation assumes a load bound for all 616 poses in the larger source family.

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